The empirical recurrence for OEIS A253867, recovered from its tail

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Abstract

OEIS A253867 records "Empirical recurrence of order 80" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 438$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 6$, so the recurrence is proved for every $n \geq 86$. Its last coefficient is $c_{80} = -136112946768375385385349842972707284 5824 \neq 0$, so no further tail satisfies anything shorter; any order-80 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A253867 is "Number of (n+2)X(4+2) 0..1 arrays with every 3X3 subblock diagonal maximum minus antidiagonal minimum nondecreasing horizontally, vertically and ne-to-sw antidiagonally.". It has offset 1 and begins

$$119040, 2546432, 52394496, 1052857184, 21171373312, 423305690976, 8407169495552, 166630395772512, \ldots$$

The entry states:

> Empirical recurrence of order 80 (see link above)

As of the "Last modified" line on the live entry (Jul 23 2025 14:19:12, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \;=\; \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 438$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 438$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \ldots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. □

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 80: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 6$ is the first whose minimal order is exactly the stated 80.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 876$ exact terms from $n = 6$ on, it returns order 80 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{80} c_i\, a(n-i),$$

c_1	31	c_{21}	12793321869793144832	c_{41}	−58916167538254616966646443540
c_2	−198	c_{22}	−8266464029936304128	c_{42}	−379501391890403086713361782538
c_3	−1714	c_{23}	−187845730407503626240	c_{43}	−627982596764546313988330435379
c_4	38299	c_{24}	−1126188983232912949248	c_{44}	−417941647569457074245044199751
c_5	−277157	c_{25}	−4629780338895543336960	c_{45}	−268637587254766349832120560517
c_6	1595804	c_{26}	−7849694176851374637056	c_{46}	4853159128564556979554420135034
c_7	−38273240	c_{27}	−71275867465526215180288	c_{47}	41046689387292876225557891184066
c_8	108280272	c_{28}	−515452721149433798983680	c_{48}	15683778505944634155284044829425
c_9	994518800	c_{29}	−1484762101668529351688192	c_{49}	21805878856425746952641168099246
c_{10}	245623104	c_{30}	−7184385538376543263260672	c_{50}	−6597978027929741023158881364423
c_{11}	99460259712	c_{31}	−33069068900988860074295296	c_{51}	−32682847176826060627383604583648
c_{12}	−580588986752	c_{32}	−71732430681970201059131392	c_{52}	−44899170812673601259109964792306
c_{13}	5540384937344	c_{33}	−167137583366588140152160256	c_{53}	282493026551523784240353015102439
c_{14}	−21631796324352	c_{34}	500662132690566185870163968	c_{54}	240773808294899721875825356856046
c_{15}	−74050054533632	c_{35}	8023390845782916852134969344	c_{55}	453805467306417326976089316778416
c_{16}	−1945491444799488	c_{36}	37733656612770723453835673600	c_{56}	−16171273559650901514004716955317
c_{17}	−14534792343533568	c_{37}	125194161915295660319147294720	c_{57}	−150797081885483107212602974376925
c_{18}	12838546109161472	c_{38}	301100401367136605049651724288	c_{58}	−249273911943055067103539634476291
c_{19}	503276329014829056	c_{39}	804618942093671017325120192512	c_{59}	−81609548720368918647502621093396
c_{20}	3737747278411132928	c_{40}	1705368816442833755126611574784	c_{60}	37268711008675201100358818611165193

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 86$, and it is the recurrence OEIS A253867 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 6$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{80} - c_1 x^{80-1} - \cdots - c_{80}$. Its constant term is $-c_{80} = 1361129467683753853853498429727072845824,$

which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 80 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 80, so $\deg q = 80$, and it is monic. It holds from some index t on. From $\max(t, 6)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 80, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 13 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 80, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is $-136112946768375385385349842972707284 5824$.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-80 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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